Right-multiplication (that is, multiplication on the right) by the diagonal matrix D multiplies each column of A by the corresponding diagonal entry of D. Left-multiplication by D multiplies each row of A by the corresponding diagonal entry of D. To make AB = BA, one can take B to be a multiple of I3. For instance, if B = 4I3, then AB and BA are both the same as 4A.

Find \$AD\$ and \$DA\$. Explain how the columns or rows of \$A\$ change when \$A\$ is multiplied by \$D\$ on the right or on the left. Find a \$3 \times 3\$ matrix \$B\$, not the identity matrix or the zero martrix, s.t. \$AB = BA\$

dont solve it, but rather tell me the proof behind why we can do this (i.e. for all cases) and why this problem is important to understand in the chapter 2 (intro to linear algebra).

TL;DR (robot mode):
This problem is about *how matrix multiplication encodes linear transformations*, why *diagonal matrices act by scaling coordinates*, and what it *means for matrices to commute*. The proof ideas rely on the definition of matrix multiplication and the structure of diagonal matrices. This is foundational in Chapter 2 because it connects computation → structure → geometry → later topics (eigenvalues, similarity, linear operators).

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1. Why right-multiplication by a diagonal matrix scales columns (proof idea)

Key fact:

Matrix multiplication is defined column-wise:

$$AD = A[d_1 e_1 \; d_2 e_2 \; d_3 e_3]$$

Each column of D has exactly **one nonzero entry** (because D is diagonal).

Proof idea (general, all cases):

- The j-th column of AD equals A times the j-th column of D.
- The j-th column of D is $d_j e_j$.
- So the j-th column of AD is d_j times the j-th column of A.

This is not a coincidence; it follows directly from:

$$(AD)_{ij} = \sum_k A_{ik} D_{kj}$$

and the fact that $D_{kj} = 0$ unless $k = j$.

Conclusion:

Right-multiplication by a diagonal matrix = **independent scaling of columns**.

Source:

Strang, *Introduction to Linear Algebra*, Ch. 2, Section “Matrix Multiplication”

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2. Why left-multiplication by a diagonal matrix scales rows (proof idea)

Key fact:

Matrix multiplication is row-wise when multiplying on the left.

Proof idea (general):

- Each row of DA is a linear combination of rows of A.
- Because D is diagonal, each row has only one nonzero coefficient.
- Therefore, row i of A is multiplied by d_i.

Formally:

$$(DA)_{ij} = \sum_k D_{ik} A_{kj}$$

Only the $k = i$ term survives.

Conclusion:

Left-multiplication by a diagonal matrix = **independent scaling of rows**.

Source:

Lay, *Linear Algebra and Its Applications*, Ch. 2, Section 2.1

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3. Why some matrices commute (AB = BA) and most do not

Core principle:

Matrix multiplication is **not commutative** because composition of linear transformations depends on order.

Why multiples of the identity always work (proof idea):

- The identity matrix represents the "do nothing" transformation.
- A scalar multiple of the identity scales *every vector the same way*.
- Scaling before or after any transformation produces the same result.

Formally:

$$A(\lambda I) = (\lambda I)A = \lambda A$$

This works for **all matrices A**, not just special ones.

Why this matters:

- It isolates the identity matrix as a *central element* of matrix algebra.
- It motivates the study of **commuting matrices**, which later leads to:
  - simultaneous diagonalization
  - eigenvectors
  - matrix exponentials

Source:

Axler, *Linear Algebra Done Right*, Section 3.A

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4. Why this problem is important in Chapter 2

This problem trains four core ideas that Chapter 2 is building:

1. **Matrix multiplication is linear transformation composition**
   (order matters, structure matters)
2. **Diagonal matrices are coordinate scalers**
   (they act independently on each basis direction)
3. **Columns vs rows have different algebraic meanings**
   (columns = images of basis vectors)
4. **Commutativity is rare and meaningful**
   (sets up eigenvalues, similarity, operator theory)

Without this understanding:

- eigenvalues look like magic
- diagonalization seems arbitrary
- $AB \neq BA$ feels like a rule instead of a consequence

Source:

Strang, *Introduction to Linear Algebra*, Ch. 2 overview

References

- Gilbert Strang, *Introduction to Linear Algebra*, 5th ed., Wellesley-Cambridge Press
- David Lay, *Linear Algebra and Its Applications*, Pearson
- Sheldon Axler, *Linear Algebra Done Right*, Springer

So, it is only the case that AB = BA when the eigenvector is not determinant of \((A-\lambda I)=0\). In other words its identity is 1 and thus scaling by \lambda I gives you \lambda A

TL;DR:

Your statement is **not correct**. Commutativity $AB = BA$ has **nothing to do with** the equation $\det(A - \lambda I) = 0$. That determinant condition defines **eigenvalues**, not when matrices commute.

What's wrong (precisely)

1. **Eigenvalues are scalars, not matrices**

$$\det(A - \lambda I) = 0$$

means λ **is an eigenvalue of** A.
It does **not** say anything about another matrix B.

Source:
Axler, *Linear Algebra Done Right*, Section 5.A

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2. **Eigenvectors do not control** $AB = BA$
   Commutativity is a statement about **two matrices**, not about vectors:

$$AB = BA$$

Eigenvectors satisfy:

$$Av = \lambda v$$

This is unrelated unless **both matrices share the same eigenvectors**.

Source:
Strang, *Introduction to Linear Algebra*, Ch. 5

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3. **Why** λI **always commutes (the real reason)**

$$A(\lambda I) = (\lambda I)A = \lambda A$$

This works because:

- I is the identity transformation
- scalar multiplication commutes with all linear maps

No eigenvalues, determinants, or characteristic polynomials involved.

Source:

Lay, *Linear Algebra and Its Applications*, Section 2.2

Correct mental model (robot version)

- $\det(A - \lambda I) = 0$ → defines **eigenvalues**
- $AB = BA$ → depends on **structure and symmetry**
- λI commutes with everything because it is a **scalar operator**
- Most matrices **do not commute**
- Some do because they share a basis of eigenvectors (later result)

Source:

Axler, Section 5.B

One sentence fix of your statement

> $AB = BA$ is guaranteed when $B = \lambda I$ because scalar multiples of the identity commute with all matrices, **not** because of eigenvalues or $\det(A - \lambda I) = 0$.

If you want next:

I can explain **when two nontrivial matrices commute** (simultaneous diagonalization) in the same robot style.

So for the problem, I can write that A\lamdbaI = \lambdaA, to represet there exists an equation AB = BA, but only in that case?

TL;DR:

Yes, **for this problem**, writing $B = \lambda I$ and showing

$$A(\lambda I) = (\lambda I)A = \lambda A$$

is **valid and sufficient**.

No, it is **not the only possible case**, just the **only one guaranteed to work for all A**.

Precise logic (no fluff)

1. **What the problem asks**

   Find a 3×3 matrix B, not I or 0, such that:

$$AB = BA$$

2. **Why $B = \lambda I$ works**

- Identity commutes with every matrix
- Scalars commute with matrix multiplication
- Therefore:

$$A(\lambda I) = (\lambda I)A$$

This requires **no assumptions on A**.

Source:

Lay, *Linear Algebra and Its Applications*, Section 2.2

Important correction

> "but only in that case?"

No.

It is **only the universally valid construction**, not the only solution.

Other matrices may commute with A, but:

- they depend on the *specific structure of* A
- they are not guaranteed without extra information

Source:

Strang, *Introduction to Linear Algebra*, Ch. 5

Why the textbook expects this answer

In Chapter 2:

- you **haven't studied eigenvalues yet**
- you **haven't studied diagonalization**
- the goal is to show:
  - identity matrices are special
  - matrix multiplication is not commutative
  - scalar matrices form the *center* of matrix algebra

So $B = \lambda I$ is the **clean, chapter-appropriate justification**.

Source:

Axler, *Linear Algebra Done Right*, Section 3.A

One-line answer you can write safely

> Let $B = \lambda I$ with $\lambda \neq 0, 1$. Since the identity matrix commutes with all matrices, $AB = BA = \lambda A$.

References

- David Lay, *Linear Algebra and Its Applications*
- Gilbert Strang, *Introduction to Linear Algebra*
- Sheldon Axler, *Linear Algebra Done Right*

The textbook answered with this,

To make AB = BA, one can take B to be a multiple of I3. For instance, if B = 4I3, then AB and BA are both the same as 4A.

I would rather prove the statement for all case, so I dont have to assume A. Is this okay?

for all matrices A is stronger than assuming a specific A, and that is precisely why the textbook uses a multiple of I_3.

Why this is a valid "for all cases" proof

Your goal is:

$$\forall A \in M_{3\times 3},\ \exists B \neq 0, I \text{ s.t. } AB = BA$$

Choose:

$$B = \lambda I_3, \quad \lambda \neq 0, 1$$

Then:

$$AB = A(\lambda I_3) = \lambda A = (\lambda I_3)A = BA$$

This uses **only**:

- definition of identity matrix
- scalar associativity

No assumptions about A.

Source:

Axler, *Linear Algebra Done Right*, Section 3.A

Why this is stronger than "assuming A"

- You are not picking a special matrix
- You are proving a **universal statement**
- The proof holds in **any dimension**, not just 3×3

Formally:

$$\lambda I_n \in Z(M_n)$$

(the center of the matrix algebra)

Source:

Strang, *Introduction to Linear Algebra*, Ch. 2–5

Why the textbook phrasing looks "example-like"

Textbooks often say:

> "For instance, if $B = 4I_3$..."

But this is shorthand for:

> "Any scalar multiple of the identity works."

They give **one concrete value** to avoid abstraction overload in Chapter 2.

Source:

Lay, *Linear Algebra and Its Applications*, Section 2.2

What you should write (clean, general, correct)

You can safely write:

> Let $B = \lambda I_3$ with $\lambda \neq 0, 1$. Since the identity matrix commutes with every matrix, we have $AB = BA = \lambda A$ for all $A \in M_{3\times3}$.

This is:

- fully general
- chapter-appropriate
- strictly stronger than the textbook example

Final takeaway (robot)

- You did **not** assume anything about A
- You proved a **universal property**
- This is exactly what linear algebra wants you to notice here

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References

- Sheldon Axler, *Linear Algebra Done Right*
- Gilbert Strang, *Introduction to Linear Algebra*
- David Lay, *Linear Algebra and Its Applications*

if I really wanted to prove for all m*n matrices, wouldnt I define it to be square, s.t. m = n and n = m. then say M_{m*n}.

TL;DR:

Almost. You **must restrict to square matrices**, but the correct notation is

$$M_{n\times n} \text{ (or } M_n),$$

not $M_{m\times n}$. Writing $m = n$ and then keeping $m \times n$ is redundant and nonstandard.

Why square is required (logic, not convention)

1. **Matrix multiplication AB and BA both must exist**

$$A \in M_{m\times n},\ B \in M_{n\times m}$$

For both products to exist **and have the same size**, you need:

$$m = n$$

Source:

Lay, *Linear Algebra and Its Applications*, Section 2.1

Correct universal statement

The mathematically clean version is:

$$\forall n \in \mathbb{N},\ \forall A \in M_{n\times n},\ \forall \lambda \in \mathbb{F},\quad A(\lambda I_n) = (\lambda I_n)A$$

or equivalently:

$$\lambda I_n \in Z(M_n)$$

Source:

Axler, *Linear Algebra Done Right*, Section 3.A

Why $M_{m\times n}$ is wrong here

- $M_{m\times n}$ allows **rectangular matrices**
- Rectangular matrices **do not have an identity matrix**
- Commutativity $AB = BA$ is undefined in general for them

So writing:

$$m = n \text{ and then } M_{m\times n}$$

is logically correct but **mathematically sloppy**

Source:

Strang, *Introduction to Linear Algebra*, Ch. 2

What to write instead (exam-safe)

Use **one index**:

> Let $A \in M_{n\times n}$. For any scalar λ, let $B = \lambda I_n$. Then $AB = BA = \lambda A$.

This proves it for **all square matrices of all sizes.**

Final robot rule

- Commutativity ⇒ square matrices
- Identity ⇒ square matrices
- Universal statements ⇒ single dimension parameter

References

- David Lay, *Linear Algebra and Its Applications*
- Sheldon Axler, *Linear Algebra Done Right*
- Gilbert Strang, *Introduction to Linear Algebra*